R16

Code No: 136FC

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester (Special) Examinations, January/February - 2021
FUNDAMENTALS OF ROBOTICS
(Computer Science and Engineering)

Time: 2 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Classify the robot anatomy according to its geometrical configurations.
b) Discuss about the working and range of any two tactile sensor. [7+8]

2.a) Describe the dynamic characteristics of a robotic arm.
b) What principle is followed for a torque sensor? [8+7]

3.a) Discuss about gear and rack method of actuating the gripper.
b) What aspects are considered in gripper selection and design? [7+8]

4.a) What are the advantages and disadvantages of actuating a magnetic gripper.
b) How the physical support is provided for the end effector? [7+8]

5. Formulate the differential kinematic equations for planar serial robot. [15]

6. Describe the DH method to provide the solution of forward and inverse kinematic problem of any standard robot. Assume the joint rotations according to the standard robot chosen. [15]

7. Express the calculations involved in computing corrected torque control. [15]

8. How lead through programming methods and teach pendant methods are useful in robot applications. [15]

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